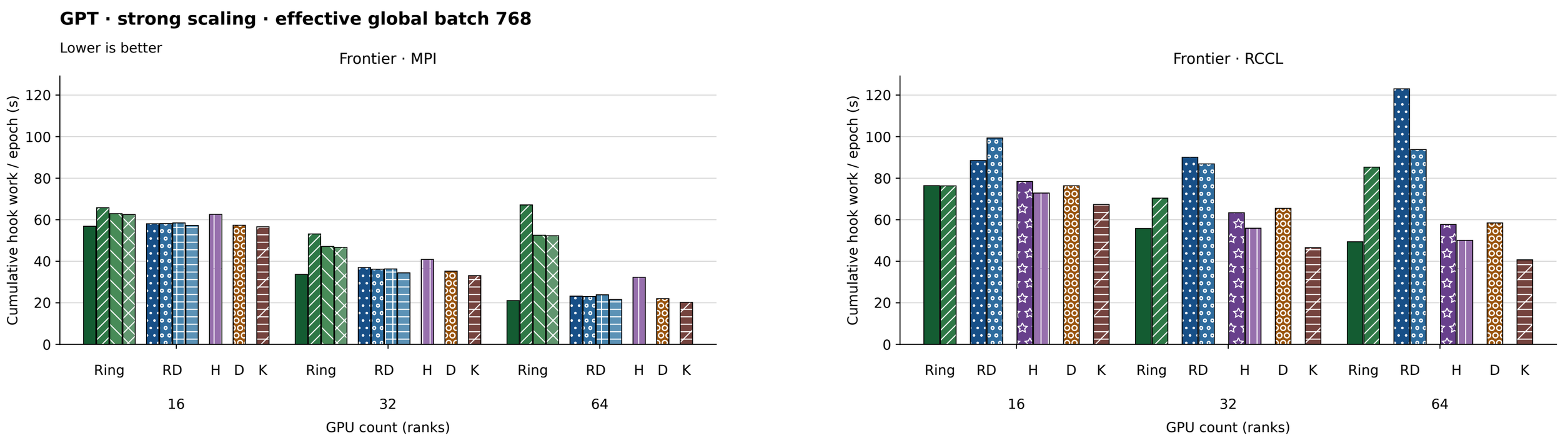

GPT · strong scaling · effective global batch 768
Lower is better
Frontier · MPI
Frontier · RCCL
Cumulative hook work / epoch (s)
0
20
40
60
80
100
120
Ring
RD
H
D
K
16
32
64
GPU count (ranks)
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Hierarchical
ZFP online · rate16
ZFP online · rate8
Direct
t = GPU count
k-ring
k = 8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate